

A Packing of 11 Four-Spaces in $\mathbb{R}^{12}$

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Abstract

We give 11 4-dimensional subspaces of $\mathbb{R}^{12}$ whose smallest squared chordal distance is at least 2.933071. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.908758, obtained by deleting 1 of the 12 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 11 subspaces of dimension 4 in $\mathbb{R}^{12}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.933071 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane’s *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 12$, $n = 4$. The cell $N = 11$ is blank in Sloane’s table, in Sloane’s coordinate files and in Cohn’s table as checked on October 8, 2026. The best value implied by known packings is 2.908758, obtained by deleting 1 of the 12 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(12 - 4)/12 \cdot 11/10 = 2.933333$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m12n4N11.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author’s laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 63 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
146653597924938788567599906373851235551378447658758881743768793499267385
391129386767827133828926570112878025436276899230476311855193362518252052
4244748373584823710290669809081300929762106445593
```

denominator:

```
4999999999999491574064720440219901149539028694883585932751938718557647135
219915454440480287705868485419921592866368930466477661203046369747517035
384977969162109082896426653855021116970285229996
```

B Coordinates

11 blocks of 12 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m12n4N11.txt`.

```
222493665496 -641428379425 -110625129978 -117699441428046238481 -445606637927 124825308626 -778841509350456839185 370092252351 -35015845549 150213243094
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